Benjamin V. Lehmann

Einstein Fellow, NASA Hubble Fellowship Program, 2026–2029. [tinyurl.com/bvlepf]
Pappalardo Fellow, MIT Center for Theoretical Physics, 2022–2025. [tinyurl.com/bvlppf]

Department of Physics & Kavli Institute for Astrophysics and Space Research
Massachusetts Institute of Technology.

(Updated October 14, 2026)

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Education

Ph. D., Physics. *Advisor:* Stefano Profumo. August 2022
University of California, Santa Cruz.

M. S., Mathematics. *Advisor:* Eleny Ionel. June 2016
Stanford University.

B. S., Mathematics; B. S., Physics, with honors. *Advisor:* Risa Wechsler. June 2015
Stanford University.

Selected awards

Michelson Postdoctoral Prize Lectureship. 2026
A competitive prize awarded to one postdoctoral scholar across all fields of physics.

Infinite Expansion Award, Massachusetts Institute of Technology. 2025
An award presented by the School of Science to "extraordinary members of the MIT community."

Josephine de Karman Dissertation Fellowship. 2021–2022
A highly competitive dissertation-year fellowship open to applicants from all fields.

Honorable Mention, Ford Foundation Dissertation Fellowship competition. March 2021
Ford Foundation & National Academies of Science, Engineering, and Medicine.

Steck Family Prize for Graduate Research Excellence, UC Santa Cruz. April 2019

ARCS Scholarship. 2018–2019
Only the department's best student is eligible for nomination for this significant award.

Marilyn Stevens Memorial Award, UC Santa Cruz Department of Physics. June 2018
This award is granted for outstanding departmental service in addition to academic excellence.

Fridley Scholarship Award, UC Santa Cruz. May 2017
The Fridley Scholarship is awarded to the single best student in the physical sciences at UCSC.

Regents' Fellowship, UC Santa Cruz. September 2016–September 2017

Seminars

Seminar, University of Tennessee, Knoxville. October 2026
Broadening direct searches for light dark matter.

Seminar, ICTP-AP. September 2026
Broadening direct searches for light dark matter.

Seminar, Brookhaven National Laboratory. July 2026
Broadening direct searches for light dark matter.

Colloquium, Case Western Reserve University. April 2026
Testing the Particle Dark Matter Paradigm.

Seminar, Case Western Reserve University. April 2026
Compact objects as dark matter: Endgame.

Seminar, Case Western Reserve University. April 2026
Broadening direct searches for light dark matter.

Seminar, Harvard University. February 2026
Broadening direct searches for light dark matter.

Seminar, Technion – Israel Institute of Technology, Israel. December 2025
New tools for dark matter physics.

Seminar, University of Michigan. November 2025
Broadening direct searches for light dark matter.

Seminar, IFAE / Universitat Autònoma de Barcelona. October 2025
Broadening direct searches for light dark matter.

KCTP Seminar, University of Chicago. May 2025
Broadening direct searches for light dark matter.

Seminar, Rutgers University New High Energy Theory Center. March 2025
New tools for dark matter physics.

Theory Seminar, SLAC. February 2025
New tools for dark matter physics.

Seminar, Boston University. February 2025
Finding primordial black holes wherever they may hide.

Seminar, Rutgers University New High Energy Theory Center. February 2025
Finding primordial black holes wherever they may hide.

Seminar, Kavli Institute for Particle Astrophysics and Cosmology. November 2024
Finding primordial black holes wherever they may hide.

IDEA Seminar, Brown University. November 2024
Finding primordial black holes wherever they may hide.

Theory Seminar, Washington University. November 2024
Finding primordial black holes wherever they may hide.

Seminar, Oklahoma State University. October 2024
Broadening direct searches for light dark matter.

Seminar, University of Oklahoma. October 2024
Broadening direct searches for light dark matter.

SOTU Seminar, Tata Institute of Fundamental Research. October 2024
Broadening direct searches for light dark matter.

Seminar, University of California, Santa Cruz. April 2024
Direct detection of dark matter far from the weak scale.

Seminar, Texas A&M University. March 2024
Direct detection of dark matter far from the weak scale.

Seminar, Queen's University. February 2024
Direct detection of dark matter far from the weak scale.

Seminar, Dark Matter: First Light Workshop, Perimeter Institute for Theoretical Physics. February 2024
Kinetic recoupling of dark matter.

Seminar, Kavli Institute for the Physics and Mathematics of the Universe. January 2024
Direct detection of dark matter far from the weak scale.

Seminar, DMPNP Workshop, Asia-Pacific Center for Theoretical Physics. January 2024
New directions for direct detection with dielectrics.

Seminar, University of Florida. November 2023
New directions for direct detection with dielectrics.

Seminar, University of Illinois Urbana-Champaign. March 2023
Kinetic recoupling of dark matter.

KICP Seminar, University of Chicago. March 2023
New frontiers in dark matter direct detection.

Seminar, Fermilab Cosmic Physics Center. March 2023
Kinetic recoupling of dark matter.

HETG Family Meeting Seminar, Harvard University. March 2023
Developments in Directional DM detection in dielectrics.

Program Seminar, Simons Center for Geometry and Physics. February 2023
Directional DM detection in dielectrics. [tinyurl.com/bddjxdbr]

Seminar, University of Texas at Austin. February 2023
Kinetic recoupling of dark matter.

HETG Seminar, Harvard University. January 2023
Kinetic recoupling of dark matter.

MIT/Tufts Cosmology Seminar December 2022
Kinetic recoupling of dark matter.

Seminar, Cornell University. November 2022
Direct detection of dark matter far from the weak scale.

Seminar, McGill University, Canada. October 2022
Direct detection of dark matter far from the weak scale.

Seminar, California Institute of Technology. October 2022
Direct detection of dark matter far from the weak scale.

Seminar, K. N. Toosi University of Technology, Iran,*. July 2022
Primordial black holes and extrasolar systems.

Seminar, Weizmann Institute of Science, Israel. May 2022
New physics from black holes with present-day tools.

Seminar, Technion – Israel Institute of Technology, Israel. May 2022
Direct detection of dark matter far from the weak scale.

Particle Physics Joint Seminar, Hebrew University, Israel. April 2022
Direct detection of dark matter far from the weak scale.

EPT Seminar, University of Maryland. February 2022
Direct detection of dark matter far from the weak scale.

Theory Colloquium, CU Boulder. February 2022
Direct detection of dark matter far from the weak scale.

Seminar, BSM PANDEMIC Series. December 2021
Massive black hole binaries as particle physics laboratories.

KIPAC Tea Talk, Stanford University. November 2021
Massive black holes as particle physics laboratories.

HEP Seminar, UC San Diego. September 2021
Extracting new physics from black holes with present-day tools.

4D Seminar, Berkeley Center for Theoretical Physics. September 2021
Direct detection of dark matter far from the weak scale.

Seminar, NANOGrav Astrophysics Working Group Meeting. June 2021
Detecting new forces in the gravitational wave background.

Seminar, SuperCDMS Science Meeting. April 2021
Dark matter–electron scattering and the dielectric function.

Seminar, Fermilab Cosmic Physics Center. March 2021
Discovering new physics in the era of black holes. [tinyurl.com/e36mb94w]

HEP Seminar, University of Notre Dame. October 2020
New opportunities for primordial black holes across the scales.

Joint UC Santa Cruz–UC Irvine theory seminar May 2020
FIMP dark matter at the KOTO experiment.

SCIPP Seminar, Santa Cruz Institute for Particle Physics. January 2020
Catching black holes in the lab.

Cosmology Seminar, UC Santa Cruz. June 2016
Some Assembly Required: Modeling Assembly Bias with Abundance Matching.

Selected conference presentations

Contributed talk, Hubble Fellows Symposium. October 2026
X-ray femtolensing probes asteroid-mass black holes.

Contributed talk, MITP Program: The Phenomenology of Primordial Black Holes. July 2026
X-ray femtolensing probes asteroid-mass black holes.

Contributed talk, Black Holes & Cosmology Workshop. June 2026
X-ray femtolensing probes asteroid-mass black holes.

Invited overview, particle dark matter mini-symposium, APS Global Physics Summit. March 2026
The Present and Future of Particle Dark Matter.

Invited talk, Rainer Weiss Memorial Symposium. February 2026
The final frontier for black holes as dark matter.

Invited plenary, SEARCH25, CERN. October 2025
Theory opportunities—is there any room for WIMPs?

Workshop talk, Aspen Center for Physics. August 2025
Physics from gravitational wave propagation.

Contributed talk, Dark Side of the Universe. July 2025
New dynamical probes of primordial black holes.

Contributed talk, Aspen Winter Conference. March 2025
Direct detection: Materials physics for discovery.

Invited talk, Miami Conference. December 2024
Broadening direct searches for light dark matter.

Contributed talk, TeVPA. August 2024
Broadening direct searches for light dark matter.

Contributed talk, APS DPF / Phenomenology Symposium. May 2024
Close encounters of the primordial kind:
a new observable for primordial black holes as dark matter.

Contributed talk, Black Holes and Cosmology Workshop. March 2024
Detecting new forces in the gravitational wave background.

Contributed talk, PLANCK. May 2023
Directional detection of dark matter with anisotropic response functions.

Contributed talk, COSMO'22. August 2022
Direct detection with superconducting nanowires.

Contributed talk, Dark Sectors of Astroparticle Physics (AstroDark-2021). December 2021
Detecting new forces in the gravitational wave background.

Contributed talk, APS Division of Particles and Fields Meeting (DPF). July 2021
Detecting new forces in the gravitational wave background.

Contributed talk, Cosmology From Home. July 2021
Detecting new forces in the gravitational wave background. [youtu.be/iSQD4i-sjMA]

Contributed talk, Phenomenology 2021 Symposium (Pheno). May 2021
Detecting new forces in the gravitational wave background.

Contributed talk, PPC. May 2021
Detecting new forces in the gravitational wave background.

Poster, "A Rainbow of Dark Sectors" Conference (Aspen Center for Physics). March 2021
Detecting new forces in the stochastic gravitational wave background.

Contributed talk, International Conference on New Frontiers in Physics (ICNFP). October 2020
FIMP dark matter at the KOTO experiment.

Contributed talk, TASI student talks. June 2020
Direct detection of primordial black hole dark matter.

Contributed talk, Phenomenology Symposium (Pheno). May 2020
FIMP dark matter at the KOTO experiment.

Contributed talk, SUSY. May 2019
Dark sources of cosmic ray spikes on Earth's doorstep.

Contributed talk, Phenomenology Symposium (Pheno). May 2019
Direct detection of primordial black hole dark matter.

Contributed talk, SUSY. July 2018
Sub-MeV Dark Matter Interactions with Electrons: Cosmological Constraints and Detection Prospects

Contributed talk, Phenomenology Symposium (Pheno). May 2018
Effective field theory and cosmology of sub-MeV leptophilic dark matter.

Poster, SLAC Summer Institutes (SSI), SLAC. August 2017
Cosmological constraints and detection prospects for sub-MeV dark matter.

Selected outreach

Volunteer instructor, The Educational Justice Institute at MIT [teji.mit.edu]. October 2022–present
Volunteer instructor in the Massachusetts correctional system.

Volunteer participant, Skype a Scientist [skypeascientist.com]. October 2019–present
This program connects scientists with classrooms for live conversations. I have spoken with 13 groups.

Physics Collaborative facilitator, UC Santa Cruz Department of Physics. January 2019–June 2022
Mentor for students in introductory physics courses.

SPS/WiPA program mentor, UC Santa Cruz Department of Physics. January 2018–June 2022
Program matches undergraduates with graduate students for long-term mentoring.

Volunteer instructor, UCSC Project for Inmate Education (PIE). October 2016–June 2022
Volunteer instructor for mathematics and astronomy courses at Santa Cruz County Jail.

Peer mentor, SU(5) [su5.group]. August 2020–2021
SU(5) was a peer mentorship organization supporting incoming physics graduate students.

Volunteer instructor. August 2015–December 2016
Prison University Project / Mount Tamalpais College [mttamcollege.edu].
Volunteer instructor for mathematics courses in a degree program at San Quentin State Prison.

Media

Scientific expert for *Science News Explores* magazine [snexplores.org]. 2025–present

Scientific advisor for PBS NOVA series *Decoding the Universe*. 2024

Students and mentoring

Current students

I am currently supervising part or all of the research work of the following students:

Joheen Chakraborty (Ph. D., MIT);
Guy Daniel Hadas (Ph. D., Hebrew U.);
Omer Katz (Ph. D., Tel Aviv U.);
Jackie Lodman (Ph. D., Harvard U.);
Rotem Ovadia (Ph. D., Hebrew U.);
Christina Schwemmbauer (Ph. D., DESY);

Undergraduate theses supervised

I directly supervised the following undergraduate theses under the oversight of my Ph. D. advisor.

Samuel D. English, January 2020–June 2022.
B. S., UC Santa Cruz ⟶ Ph. D. (Physics), University of New Mexico.
The Hidden Friend's Wake: Dark Matter and a Binary at the Galactic Center.
Selected for **Honors** in the major.

Olivia G. Ross, November 2018–June 2021.
B. S., UC Santa Cruz ⟶ Ph. D. (Astronomy), Cornell University.
Searching for primordial black holes.
Selected for **Honors** in the major.
Steck Family Award for the single best senior thesis of 2020–2021 at UC Santa Cruz.

Paul Andreini, March 2016–September 2019.
B. S., UC Santa Cruz ⟶ Ph. D. (Physics), University of Oregon.
Time-evolution of the primordial black hole mass function.
Selected for **Honors** in the major.

Thomas Schwemberger, February 2018–December 2018.
B. S., UC Santa Cruz ⟶ Ph. D. (Physics), University of Oregon.
Constraints on Planck mass relics from primordial black hole evaporation with a generic initial mass function.
Selected for **Honors** in the major.

Anikeya Aditya, September 2016–June 2018.
B. S., UC Santa Cruz ⟶ Ph. D. (Materials Science), University of Southern California.
Inferring the mass function of primordial black holes from gravitational wave observations.
Selected for **Honors** in the major.

Other former student collaborators

Badal Bhalla, October 2022–May 2026.
Ph. D., University of Oklahoma ⟶ Postdoctoral Associate, University of Utah.

Bethany Suter, May 2023–May 2026.
Ph. D., UC Berkeley ⟶ Postdoctoral Associate, Fermilab.

Marcell Howard, January 2024–May 2026.
Ph. D., University of Pittsburgh ⟶ Postdoctoral Associate, Brown University.

Cara Giovanetti, January 2024–May 2025.
Ph. D., NYU ⟶ Postdoctoral Fellow, UC Berkeley.

Wenzer Qin, November 2023–May 2024.
Ph. D., MIT ⟶ Simons Junior Fellow, NYU.

Sarah R. Geller, January 2023–May 2024.
Ph. D., MIT ⟶ NSF MPS Postdoctoral Fellow, UC Santa Cruz / UC Berkeley.

Tung X. Tran, January 2023–May 2024.
B. S., MIT ⟶ Ph. D. (Physics), Stanford University.

Nolan Smyth, January 2021–June 2024.
Ph. D., UC Santa Cruz ⟶ Postdoctoral Fellow, Mila / Université de Montréal.

Ava Webber, September 2020–June 2022.
B. S., UC Santa Cruz ⟶ Ph. D. (Astronomy), Clemson University.

Mason Hargrave, September 2018–June 2019.
B. S., UC Santa Cruz ⟶ Ph. D. (Computational Biology), Rockefeller University.

Jackson Yant, January 2016–June 2018.
B. S., UC Santa Cruz ⟶ Ph. D. (Physics), Dartmouth College.

Professional activities

Referee for Physical Review Letters (**PRL**), Physical Review D (**PRD**), Journal of High Energy Physics (**JHEP**), Journal of Cosmology and Astroparticle Physics (**JCAP**), European Physical Journal C (**EPJC**), The Astrophysical Journal (**ApJ**), Astrophysics and Space Science (**Ap&SS**), and Physics of the Dark Universe (**PDU**).

Subject expert for Berggruen Prize Essay Competition. 2025

Scientific organizing committee member, Hubble Fellows Symposium. 2026

Other workshops and schools

Aspen Summer Workshop: *Physics in the Era of Nanohertz Gravitational Waves.* August 2025
Aspen Center for Physics.

Aspen Winter Conference: *Empowering the New Vision in High Energy Physics.* March 2025
Aspen Center for Physics.

KITP Program: June–July 2024
Dark Matter Theory, Simulation, and Analysis in the Era of Large Surveys.
Kavli Institute for Theoretical Physics.

APCTP Workshop: *Dark Matter as a Portal to New Physics (DMPNP).* January 2024
Asia-Pacific Center for Theoretical Physics.

GUINEAPIG Workshop, University of Montreal. July 2023

MITP Workshop: *Theoretical Particle Cosmology in the Early and Late Universe.* May 2023
MITP & Corfu Summer Institute.

SCGP Program: March 2023
Lighting new Lampposts for Dark Matter and Beyond the Standard Model.
Simons Center for Geometry and Physics.

EXCESS2022 Workshop, Karlsruhe Institute of Technology. February 2022

Les Houches Summer School 2021: *Dark Matter.* July–August 2021
École de Physique des Houches.

TASI 2020: *The Obscure Universe: Neutrinos and other Dark Matters.* June 2020
Theoretical Advanced Study Institute.

37th Advanced School in Theoretical Physics: December 2019–January 2020
New Ideas for Old Puzzles in Particle Physics.
Israel Institute for Advanced Studies.

GGI Workshop: *Next Frontiers in the Search for Dark Matter.* August–September 2019
Galileo Galilei Institute.

45th SLAC Summer Institute: *Cosmic Opportunities.* July 2017
SLAC National Accelerator Laboratory.

Committee service

Physics Values Committee. **Co-chair,** November 2023–present. October 2022–present
Department of Physics, Massachusetts Institute of Technology.
[physvals.mit.edu]

Nuclear and Particle Theory Seminar Committee. August 2024–May 2025
Department of Physics, Massachusetts Institute of Technology.

Diversity and Climate Committee. February 2018–June 2022
Department of Physics, UC Santa Cruz.
[www.physics.ucsc.edu/about/diversity/diversity-committee.html]

Colloquium Committee. June 2018–June 2022
Department of Physics, UC Santa Cruz.

CUWiP 2022/2023 Local Organizing Committee at UC Santa Cruz. January 2020–January 2023
Conference for Undergraduate Women in Physics.

Teaching experience

Selected student comments are provided for all courses in which I received any narrative feedback. Due to platform changes, student evaluations are not available between Spring 2019 and Spring 2020.

Teaching assistant, UC Santa Cruz.

Introductory Physics II (PHYS 5B). March–June 2021
"It's difficult to explain how good of a TA Ben actually was. [...] His teaching method was spot on."

Thermodynamics and Statistical Mechanics (PHYS 112). January–March 2021
"Ben is a very kind and inclusive person, great at mentoring and teaching..."
"I really appreciated the feeling of support that Ben gave us as students..."
"He's the goat and deserves a raise forreal."

General Chemistry III (CHEM 1C). September–December 2020
"Best TA I ever had."

Particle Astrophysics (PHYS 129). January–April 2020

General Chemistry III (CHEM 1C). September–December 2019

Introductory Physics I Laboratory (PHYS 6L). April–June 2019

Computational Physics (PHYS 115). April–June 2019

Conceptual Physics (PHYS 1). January–April 2019
"Everything I learned about physics this quarter I owe to Wednesday sections with Ben."

Mathematical Methods in Physics III (PHYS 116C). September–December 2018
"The best TA ever!"

Quantum Mechanics I (PHYS 139A). April–June 2018
"He knew when to give me answers versus [...] to suggest that I struggle a little more."
"Ben's awesome."

Physics and Psychophysics of Music (PHYS / MUSC 80U). January–March 2018
"Ben lets students think for themselves."

Lead TA, Introductory Physics I (PHYS 6A). September–December 2017
"Engaging, helpful, and extraordinarily friendly."
"Ben is good at teaching in a way that is easy to understand."
"Ben was clear and taught you how to think in physics."

Introductory Physics I Laboratory (PHYS 6L). June–August 2017
"Really enjoyed the way Ben taught the lab, he was awesome!"

Introductory Physics I Laboratory (PHYS 6L). April–June 2017
"I know I can never be as patient as him even next life."
"Everyone in class loves Ben."

Introductory Physics II Laboratory (PHYS 6M). January–March 2017
"Ben is one of the best TAs I have ever had."

Introductory Physics I Laboratory (PHYS 6L). September–December 2016
"When we were completely confused, he helped point us in the right direction."

Mount Tamalpais College / Prison University Project.

Lead instructor, Intermediate Algebra (MATH 115). May–August 2016

Co-instructor, Elementary Algebra (MATH 99). January–April 2016

Study group leader, Elementary Algebra (MATH 99). August–December 2015

Teaching assistant, Stanford University.

Electricity, Magnetism, and Optics (PHYSICS 23). January–March 2014
"Because of [Ben] I am actually planning on getting a minor in physics."
"I never asked a question that Ben couldn't give a good answer to."
"Good section leader who obviously cared a lot about us."

Publications

An up-to-date list of publications can be found at http://inspirehep.net/authors/1737895.
The majority of my publications follow the HEP convention, in which author order is alphabetical.

★ I was the project leader
■ I was the senior author
† I was an equal contributor (alphabetical author order)

[1] ■ **BVL**, Jackie Lodman, and Thomas Steingasser. Tunneling and tidal stripping in multifield ultralight dark matter halos. May 2026, 2605.12589. Submitted to *JHEP*.

[2] ★ Sinéad M. Griffin, Guy Daniel Hadas, Yonit Hochberg, Katherine Inzani, and **BVL**. Dark Matter–Electron Detectors for Dark Matter–Nucleon Interactions. *Phys. Rev. Lett.*, 135(14):141803, September 2025, 2412.16283. DOI 10.1103/6qqv-rl7q.

[3] ★ Badal Bhalla, **BVL**, Kuver Sinha, and Tao Xu. Fast and Fewrious: Stochastic binary perturbations from fast compact objects. *Phys. Rev. D*, 113(4):043040, July 2025, 2507.16894. DOI 10.1103/k56g-tf9j.

[4] ★ Sinéad M. Griffin, Yonit Hochberg, **BVL**, Rotem Ovadia, Kristin A. Persson, Bethany A. Suter, Ruo Xi Yang, and Wayne Zhao. First High-Throughput Evaluation of Dark Matter Detector Materials. *Phys. Rev. Lett.*, 136(19):191801, June 2025, 2506.19905. DOI 10.1103/bb4g-7939.

[5] † Christina Schwemmbauer, **BVL**, et al. First direct search for light dark matter interactions in a transition-edge sensor. *Phys. Rev. D*, 114(1):012006, June 2025, 2506.18982. DOI 10.1103/zytz-n25v.

[6] ★ **BVL**. Higher multipoles of the cow. *Am. J. Phys.*, April 2025, 2504.00506. A tongue-in-cheek but rigorous extension of the spherical cow approximation. **Featured Article**. [**AIP Scilight**]. DOI 10.1119/5.0287411.

[7] † Laura Baudis, **BVL**, et al. A New Bite Into Dark Matter with the SNSPD-Based QROCODILE Experiment. *Phys. Rev. Lett.*, 135:081002, December 2024, 2412.16279. [**Featured in Physics**] [**ScienceDaily**]. DOI 10.1103/4hb6-f6jl.

[8] ★ Badal Bhalla, **BVL**, Kuver Sinha, and Tao Xu. Dancing with invisible partners: Three-body exchanges with primordial black holes. *Phys. Rev. D*, 111(4):043029, August 2024, 2408.04697. [**New Scientist**] [**Astrobites**] DOI 10.1103/PhysRevD.111.043029.

[9] ★ Jiansong Gao, Yonit Hochberg, **BVL**, Sae Woo Nam, Paul Szypryt, Michael R. Vissers, and Tao Xu. Detecting Light Dark Matter with Kinetic Inductance Detectors. March 2024, 2403.19739. Submitted to *Physical Review Letters*.

[10] ★ Tung X. Tran, Sarah R. Geller, **BVL**, and David I. Kaiser. Close encounters of the primordial kind: a new observable for primordial black holes as dark matter. *Phys. Rev. D*, 110(6):063533, December 2023, 2312.17217. **Editors' Suggestion**. [**Featured in Physics**] [**LA Times**] [**Astrobites**] [**SciAm**] DOI 10.1103/PhysRevD.110.063533.

[11] ★ **BVL**, Logan Morrison, Stefano Profumo, and Nolan Smyth. Kinetic recoupling of dark matter. *JCAP*, 07:049, October 2023, 2310.20513. DOI 10.1088/1475-7516/2024/07/049.

[12] ★ Yonit Hochberg, Eric David Kramer, Noah Kurinsky, and **BVL**. Directional detection of light dark matter in superconductors. *Phys. Rev. D*, 107(7):076015, September 2023, 2109.04473. DOI 10.1103/PhysRevD.107.076015.

[13] † Wolfgang Altmannshofer, Stefania Gori, **BVL**, and Jianhong Zuo. UV physics from IR features: New prospects from top flavor violation. *Phys. Rev. D*, 107(9):095025, March 2023, 2303.00781. DOI 10.1103/PhysRevD.107.095025.

[14] † Christian Boyd, Yonit Hochberg, Yonatan Kahn, Eric David Kramer, Noah Kurinsky, **BVL**, and To Chin Yu. Directional detection of dark matter with anisotropic response functions. *Phys. Rev. D*, 108(1):015015, December 2022, 2212.04505. DOI 10.1103/PhysRevD.108.015015.

[15] ★ Yonit Hochberg, **BVL**, Ilya Charaev, Jeff Chiles, Marco Colangelo, Sae Woo Nam, and Karl K. Berggren. New constraints on dark matter from superconducting nanowires. *Phys. Rev. D*, 106(11):112005, December 2022, 2110.01586. **Editors' Suggestion**. [**Featured in Physics**] DOI 10.1103/PhysRevD.106.112005.

[16] ★ **BVL** and Stefano Profumo. Black hole remnants are not too fast to be dark matter. *Phys. Dark Univ.*, 39:101145, November 2022, 2105.01627. DOI 10.1016/j.dark.2022.101145.

[17] ★ **BVL**, Ava Webber, Olivia G. Ross, and Stefano Profumo. Capture of primordial black holes in extrasolar systems. *JCAP*, 08:079, August 2022, 2205.09756. DOI 10.1088/1475-7516/2022/08/079.

[18] † Daniel Davies, Michael Dine, and **BVL**. Light Quarks at Large N. January 2022, 2201.05719. Contribution to a Festschrift for Peter Suranyi.

[19] † Jeff A. Dror, **BVL**, Hiren H. Patel, and Stefano Profumo. Discovering new forces with gravitational waves from supermassive black holes. *Physical Review D*, 104(8):083021, October 2021, 2105.04559. DOI 10.1103/PhysRevD.104.083021.

[20] † Yonit Hochberg, Yonatan Kahn, Noah Kurinsky, **BVL**, To Chin Yu, and Karl K. Berggren. Determining Dark Matter–Electron Scattering Rates from the Dielectric Function. *Phys. Rev. Lett.*, 127:151802, October 2021, 2101.08263. DOI 10.1103/PhysRevLett.127.151802.

[21] ★ **BVL**, Olivia G. Ross, Ava Webber, and Stefano Profumo. Three-body capture, ejection, and the demographics of bound objects in binary systems. *Monthly Notices of the Royal Astronomical Society*, 505(1):1017–1028, April 2021, 2012.05875. DOI 10.1093/mnras/stab1121.

[22] ★ **BVL**, Stefano Profumo, and Jackson Yant. Model-independent discovery prospects for primordial black holes at LIGO. *Monthly Notices of the Royal Astronomical Society*, 501(3):3727–3740, December 2020, 2007.00021. DOI 10.1093/mnras/staa3806.

[23] † Wolfgang Altmannshofer, **BVL**, and Stefano Profumo. Cosmological implications of the KOTO excess. *Physical Review D*, 102(8):083527, October 2020, 2006.05064. DOI 10.1103/PhysRevD.102.083527.

[24] ★ **BVL** and Stefano Profumo. Cosmology and prospects for sub-MeV dark matter in electron recoil experiments. *Physical Review D*, 102(2):023038, July 2020, 2002.07809. DOI 10.1103/PhysRevD.102.023038.

[25] † Adam Coogan, **BVL**, and Stefano Profumo. Connecting direct and indirect detection with a dark spike in the cosmic-ray electron spectrum. *Journal of Cosmology and Astroparticle Physics*, 2019(10):063–063, October 2019, 1903.07177. DOI 10.1088/1475-7516/2019/10/063.

[26] ★ **BVL**, Christian Johnson, Stefano Profumo, and Thomas Schwemberger. Direct detection of primordial black hole relics as dark matter. *Journal of Cosmology and Astroparticle Physics*, 2019(10):046–046, October 2019, 1906.06348. DOI 10.1088/1475-7516/2019/10/046.

[27] ★ **BVL**, Stefano Profumo, and Jackson Yant. The maximal-density mass function for primordial black hole dark matter. *Journal of Cosmology and Astroparticle Physics*, 2018(04):007, April 2018, 1801.00808. DOI 10.1088/1475-7516/2018/04/007.

[28] ★ **BVL**, Yao-Yuan Mao, Matthew R. Becker, Samuel W. Skillman, and Risa H. Wechsler. The Concentration Dependence of the Galaxy-Halo Connection: Modeling Assembly Bias with Abundance Matching. *The Astrophysical Journal*, 834(1), December 2016, 1510.05651. DOI 10.3847/1538-4357/834/1/37.

Proceedings

[29] Christina Schwemmbauer, Yonit Hochberg, Katharina-Sophie Isleif, Friederike Januschek, **BVL**, Axel Lindner, Manuel Meyer, Gulden Othman, and Jose Alejandro Rubiera Gimeno. Simulations for direct dark matter searches using ALPS II's TES detection system. *PoS*, COSMICWISPers:055, July 2024, 2407.02954. DOI 10.22323/1.454.0055.

[30] Christina Schwemmbauer et al. Direct dark matter searches using ALPS II's TES detection system. *PoS*, EPS-HEP2023:120, February 2024. DOI 10.22323/1.449.0120.

[31] Simeon Bird et al. Snowmass2021 Cosmic Frontier White Paper: Primordial black hole dark matter. *Phys. Dark Univ.*, 41:101231, 2023, 2203.08967. From *2022 Snowmass Summer Study*. DOI 10.1016/j.dark.2023.101231.

[32] Arka Banerjee et al. Snowmass2021 Cosmic Frontier White Paper: Cosmological Simulations for Dark Matter Physics. In *2022 Snowmass Summer Study*, March 2022, 2203.07049. 2203.07049.

[33] Masha Baryakhtar et al. Dark Matter In Extreme Astrophysical Environments. In *2022 Snowmass Summer Study*, March 2022, 2203.07984. 2203.07984.

[34] Daniel Carney et al. Snowmass2021 Cosmic Frontier White Paper: Ultraheavy particle dark matter. In *2022 Snowmass Summer Study*, March 2022, 2203.06508. 2203.06508.

References

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